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BMJ Open

Non-pharmacological interventions in primary care to improve the quality of life of older patients with palliative care needs: a systematic review protocol

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Keywords:	PALLIATIVE CARE, GERIATRIC MEDICINE, GENERAL MEDICINE (see Internal Medicine)

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Manuscripts

Title page

Title of the review

Non-pharmacological interventions in primary care to improve the quality of life of older patients with palliative care needs: a systematic review protocol

Running head

Palliative Care in Primary Care

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Abstract

Introduction:
In the last decades, the number of older people living with chronic diseases has rapidly increased. The prevalence of palliative care needs in this population can reach 17%, making the general practitioner (GP) a cornerstone in the identification and first medical intervention delivery. Therefore, knowing the primary care interventions that effectively improve the quality of life of these patients can play an important role in the delivery of health care.

Methods and analysis: We will systematically review randomized controlled trials evaluating the effect of non-pharmacologic primary care interventions on the quality of life of older patients (≥65 years) with palliative care needs. PsycInfo, EMBASE, MEDLINE, Cochrane and CINAHL will be searched until December 2021. Screening, data extraction and quality evaluation (using the Cochrane risk-of-bias tool) will be done by independently by two reviewers, with disagreements solved by a third reviewer. We will conduct meta-analysis if appropriate. In case of high heterogeneity, findings will be analysed by subgroup according to intervention type, main disease/symptoms and care context. We will perform a sensitivity analysis based on study quality. Evidence of publication bias will be assessed using funnel plots.

Ethics and dissemination: Formal ethical approval is not required as primary data will not be collected. The results will be disseminated through a peer-reviewed publication, conference presentation and the press.

Trial registration number International Prospective Register for Systematic Reviews (PROSPERO) number: CRD42020154216

Keywords: Palliative care, Older patients, Non-pharmacological interventions, Quality of life, Systematic Review, Meta-analysis.

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Article Summary

Strengths and limitations of this study

- Summarize the existing evidence on the impact that non-pharmacological interventions applicable in primary care settings can have in improving the quality of life of older adults with palliative care needs
- These findings can help to service developments and professional training, with a view to improve care and thus the quality of life of the patients and their families.
- Potential limitations of the review include the expected heterogeneity in terms of populations, interventions, care contexts and outcome measurement between the studies

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Introduction

In the last decades, the number of people living with chronic diseases has rapidly increased, mainly due to the ageing of the population, leading to a rise in dependency status and entailing important social costs. (1) These chronic, progressive, life-threatening and burdensome diseases play an important role in this new era of the palliative care approach. The prevalence of palliative care needs in older people ranges from 8 up to 17%, depending on the population studied and the tools used to identify needs. (2,3)

Palliative care is a philosophy of care focused on improving the quality of life (QoL) of patients and their family members in the process of coping with death through early identification, prevention and relief of suffering, evaluation of treatment appropriate to physical, psychosocial and spiritual problems. (4) The evaluation of QoL of patients in palliative care is an important procedure in the identification of a patient's overall condition as well as in the evaluation of the quality of service provided. (5)

The occurrence of incurable diseases can cause an enormous challenge to the patient, his/her family as well as medical professionals, affecting the patient' QoL in many ways. (6) This holistic paradigm calls for new measures to reduce suffering and provide comfort, which is the key goal of medicine, particularly in palliative care.(7)

Pharmacological techniques have improved and are now more capable of managing physical pain. However, palliative care extends beyond the relief of physical symptoms as it seeks to strengthen the psychological, spiritual and social domains to provide greater comfort to patients. (8,9) Therefore, non-pharmacological interventions had been increasingly used in palliative care to promote comfort and improve patients' satisfaction with end-of-life care. (10–12)

Primary care professionals promote a community-based care delivery and a long-lasting follow-up of their patients. These professionals are in a good position to contribute to the early identification of patients with chronic, progressive, life-threatening and burdensome diseases and to intervene in the delivery of palliative and end-of-life care. Some evidence supports that when GPs are involved in care, palliative care seems to improve, with benefits for the patient and his/her family.(13,14) However, no systematic review has been conducted on this topic, identifying the most effective interventions and therefore helping in choosing the best ones to use in the primary care setting. Thus, we aim to determine the impact of the non-pharmacological interventions used in the primary care setting, in improving the QoL of older patients with palliative care needs.

Methods and design

Identification and selection of articles

This systematic review will include randomized controlled trials - RCTs (study design) conducted with adults aged ≥65 years with palliative care needs (population) evaluating the impact on QoL (outcome) of nonpharmacological interventions (intervention) used in primary care settings.

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Patients with palliative care needs will be defined as patients diagnosed with a chronic, progressive and potentially fatal illness.

QoL will be assessed according to the score results of standardized and validated tools used to evaluate these domains (e.g. WHOQOL(16), QOLS(17), EQ-5D(18), SF-36(19), SF-12(19)).

As secondary outcomes, we will consider patients' symptoms relief (namely, pain, dyspnea, constipation, distress, depression and other important symptoms presented) and other patients' well-being measurements.

The search will be conducted in PsycInfo, EMBASE, MEDLINE, Cochrane and CINAHL from the start of indexing until December 2021. To identify ongoing clinical trials, we will also search in clinicaltrials.gov. No restriction on the language of publication will apply.

The search will be complemented by a manual search of abstract books of relevant congresses and scientific meetings in the last five years, namely the *World Congress of the European Association for Palliative Care*, *International Congress on Palliative Care, WONCA Europe Conference*. Experts in this area will also be contacted to identify pertinent articles that may have not been identified by our query. Authors of retrieved articles will be contacted in case additional clarification is required.

When analyzing the search results, we will first screen the titles and abstracts of the articles found in the search. Then, all potential candidates will be selected by reading their full-text. Both phases will be independently carried out by two reviewers. Reasons for exclusion will be recorded. Disagreements will be solved by a third reviewer. Reviewer agreement in the selection process will be evaluated using agreement proportion and Kappa statistics.

Patient and Public Involvement
No patient involved.

Data extraction

The extraction of data from the studies will be independently carried out by two reviewers, with disagreements solved by a third reviewer. Customized data extraction tables will be created, piloted with five of the included studies and used to collect relevant data from all. To conduct data extraction, it will be used a standard data extraction form.

The following data will be extracted: general characteristics of the study (study design, sample size and setting), socio-demographic sample characteristics (gender, age and care context, e.g. home, care home, hospice, palliative care unit), main disease, main symptoms, the description of the intervention and comparator, methods of QoL assessment and other reported outcomes such as symptom relief or other measures of well-being.

Whenever possible, raw data will be extracted, allowing the calculation of the effect measure (such as QoL, symptom relief or other well-being measures). When data are ambiguous or missing from the published study, the corresponding author will be contacted.

Methodological quality assessment

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We will evaluate the quality of the studies using the Cochrane risk-of-bias tool (15). Two reviewers will independently apply the tool to each study, with disagreements solved by a third reviewer.

Analysis

We will undertake a narrative synthesis of the findings from the included studies, structured around the type of interventions and target population characteristics. Each intervention will be summarized in terms of its care context, content and format, and resulting effects.

In case of high heterogeneity, subgroup analysis will be performed according to the type of intervention, main disease/main symptoms and care context. Using the RevMan v.5.1® software, we will extract the data from the primary studies and analyze to decide whether it makes sense, according to heterogeneity and number of comparable papers, to perform a quantitative synthesis through meta-analysis to obtain an aggregate measure of QoL.

Heterogeneity will be evaluated using the Cochran Q test (for a significance level of 0.05) and the I-squared statistic. In the absence of heterogeneity, we will use the Model of Fixed Effects. In the presence of slight to moderate heterogeneity (I-squared <40-50%), we will use the Random Effects Model. In case of severe heterogeneity (I-squared >40-50%), even in the subgroup analysis, which impedes the accomplishment of an adequate quantitative synthesis, we will attempt to explain the existing variability.

We will also conduct a sensitivity analysis based on study quality and the evidence of publication bias will be assessed by visual inspections of funnel plots. QoL will be evaluated comparing standard means difference.

Conclusion

This review will search, critically appraise and summarize the existing evidence on the impact that non-pharmacological interventions applicable in primary care settings can have in improving the quality of life of older adults with palliative care needs.

There are some studies assessing non-pharmacological interventions to improve the QoL of patients with palliative care needs (20-22). However, as far as we are aware, none summarizes the interventions that can be performed in primary care. Considering a holistic definition of palliative care, the involvement of primary care is essential, and its professionals need to be empowered with the best scientific evidence to deliver good quality care to their patients. By publishing the research protocol, we reinforce the clarity of the strategy and minimize the risk of bias, namely on selective outcome reporting.

Potential limitations of the review include the expected heterogeneity in terms of populations, interventions, care contexts and outcome measurement between the studies.

However, despite limitations, the review findings can be valuable and will be disseminated to help inform policy, service developments and professional training, with a view to improve care and thus the quality of life of the patients and their families.

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MEDLINE Query

Group 1

1. exp Terminal care/
2. exp Palliative care/
3. exp Terminally ill /
4. exp Hospices/
5. exp Death/
6. Palliat*.mp
7. Hospice*.mp
8. End of life.mp
9. Advanced or end stage or terminal adj4 disease* or illness*.mp
10. 1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 or 9

Group 2

11. exp Aged/
12. exp Old Age Assistance/
13. Old or aged or elderly.mp
14. 11 or 12 or 13

Group 3

15. exp Primary Health Care
16. exp General Practice
17. exp Family Practice
18. exp Community Health Centers
19. exp General Practitioners
20. exp Physicians, Family
21. exp Physicians, Primary Care
22. Primary care or family practice* or health center* or health centre*.mp
23. General practitioner* or GP* or family doctor*.mp
24. 15 or 16 or 17 or 18 or 19 or 20 or 21 or 22 or 23 or 24

Group 4

25. randomized controlled trial.pt
26. controlled clinical trial.pt
27. randomized.ti or randomized.ab
28. placebo.ti or placebo.ab
29. clinical trials as topic/
30. randomly.ti or randomly.ab
31. trial.ti
32. 25 or 26 or 27 or 28 or 29 or 30 or 31
33. animals.sh NOT humans.sh
34. 32 not 33

Group 5

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35. Exp “Quality of life“/
36. Quality of life.mp
37. 35 or 36

38. 10 and 14 and 24 and 34 and 37

To adapt to PsyncInfo, EMBASE, Medline, Cochrane and Cinahl search.

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Reporting checklist for protocol of a systematic review and meta analysis.

Based on the PRISMA-P guidelines.

Instructions to authors

Complete this checklist by entering the page numbers from your manuscript where readers will find each of the items listed below.

Your article may not currently address all the items on the checklist. Please modify your text to include the missing information. If you are certain that an item does not apply, please write "n/a" and provide a short explanation.

Upload your completed checklist as an extra file when you submit to a journal.

In your methods section, say that you used the PRISMA-Preporting guidelines, and cite them as:

Moher D, Shamseer L, Clarke M, Ghersi D, Liberati A, Petticrew M, Shekelle P, Stewart LA. Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols (PRISMA-P) 2015 statement. Syst Rev. 2015;4(1):1.

		Reporting Item	Page Number
Title			
Identification	#1a	Identify the report as a protocol of a systematic review	1
Update	#1b	If the protocol is for an update of a previous systematic review, identify as such	n/a
Registration			
	#2	If registered, provide the name of the registry (such as PROSPERO) and registration number	2
Authors			
Contact	#3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	2
Contribution	#3b	Describe contributions of protocol authors and identify the guarantor of the review	7

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Amendments			
	#4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	n/a
Support			
Sources	#5a	Indicate sources of financial or other support for the review	7
Sponsor	#5b	Provide name for the review funder and / or sponsor	n/a
Role of sponsor or funder	#5c	Describe roles of funder(s), sponsor(s), and / or institution(s), if any, in developing the protocol	n/a
Introduction			
Rationale	#6	Describe the rationale for the review in the context of what is already known	4
Objectives	#7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	2,3,4
Methods			
Eligibility criteria	#8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	2,3
Information sources	#9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	3
Search strategy	#10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	10,11
Study records - data management	#11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	2,3,4
Study records - selection process	#11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	2,3
Study records - data	#11c	Describe planned method of extracting data from reports (such as	3,4

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collection process		piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	
Data items	#12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	3,4
Outcomes and prioritization	#13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	3,4
Risk of bias in individual studies	#14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	2,3,4
Data synthesis	#15a	Describe criteria under which study data will be quantitatively synthesised	3,4
Data synthesis	#15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I2, Kendall's τ)	3,4
Data synthesis	#15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	3,4
Data synthesis	#15d	If quantitative synthesis is not appropriate, describe the type of summary planned	n/a
Meta-bias(es)	#16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	3,4
Confidence in cumulative evidence	#17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	3,4

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Keywords:	PALLIATIVE CARE, GERIATRIC MEDICINE, GENERAL MEDICINE (see Internal Medicine)

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Title page

Title of the review

Non-pharmacological interventions in primary care to improve the quality of life of older patients with palliative care needs: a systematic review protocol

Running head

Palliative Care in Primary Care

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Abstract

Introduction:
In the last decades, the number of older people living with chronic diseases has rapidly increased. The prevalence of palliative care needs in this population can reach 17%, making the general practitioner (GP) a cornerstone in the identification and first medical intervention delivery. Therefore, knowing the primary care interventions that effectively improve the quality of life of these patients can play an important role in the delivery of health care.

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Strengths and limitations of this study

- This protocol describes a comprehensive systematic review of RCTs, prospectively registered in PROSPERO and guided by PRISMA and Cochrane recommendations.
- Subgroup analyses can help with the expected heterogeneity and identify groups for whom the interventions have greater effectiveness.
- The data summarized can be valuable to develop primary palliative care.

Ethics and dissemination: Formal ethical approval is not required as primary data will not be collected. The results will be disseminated through a peer-reviewed publication, conference presentation and the press.

Trial registration number International Prospective Register for Systematic Reviews (PROSPERO) number: CRD42020154216

Keywords: Palliative care, Older patients, Non-pharmacological interventions, Quality of life, Systematic Review, Meta-analysis.

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Introduction

In the last decades, the number of people living with chronic diseases has rapidly increased, mainly due to the ageing of the population, leading to a rise in dependency status and entailing important social costs.(1) These chronic, progressive, life-threatening and burdensome diseases play an important role in this new era of the palliative care approach. The prevalence of palliative care needs in older people ranges from 8 up to 17%, depending on the population studied and the tools used to identify needs.(2,3)

Palliative care is a philosophy of care focused on improving the quality of life (QoL) of patients and their family members in the process of coping with death through early identification, prevention and relief of suffering, evaluation of treatment appropriate to physical, psychosocial and spiritual problems.(4) The evaluation of QoL of patients in palliative care is an important procedure in the identification of a patient's overall condition as well as in the evaluation of the quality of service provided.(5)

The occurrence of incurable diseases can cause an enormous challenge to the patient, his/her family as well as medical professionals, affecting the patient' QoL in many ways.(6) This holistic paradigm calls for new measures to reduce suffering and provide comfort, which is the key goal of medicine, particularly in palliative care.(7)

Pharmacological techniques have improved and are now more capable of managing physical pain. However, palliative care extends beyond the relief of physical symptoms as it seeks to strengthen the psychological, spiritual and social domains to provide greater comfort to patients. (8,9) Therefore, non-pharmacological interventions had been increasingly used in palliative care to promote comfort and improve patients' satisfaction with end-of-life care. (10–12)

Primary care professionals promote a community-based care delivery and a long-lasting follow-up of their patients. These professionals are in a good position to contribute to the early identification of patients with chronic, progressive, life-threatening and burdensome diseases and to intervene in the delivery of palliative and end-of-life care. Some evidence supports that when GPs are involved in care, palliative care seems to improve, with benefits for the patient and his/her family.(13,14) However, no systematic review has been conducted on this topic, identifying the most effective interventions and therefore helping in choosing the best ones to use in the primary care setting. Thus, we aim to determine the effectiveness of the non-pharmacological interventions used in the primary care setting, in improving the QoL of older patients with palliative care needs.

Methods and design

The systematic review was prospectively registered with PROSPERO (CRD42020154216). Reporting complies with Preferred Reporting Items for Systematic Review and Meta-Analysis guidance (PRISMA 2020).(15)

Eligibility criteria

This systematic review will include randomized controlled trials - RCTs (study design) conducted with adults aged ≥65 years with palliative care needs (population). Both patient- and cluster-randomised trials will be included. Quasi-randomised trials will be excluded.

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For a study to be included, the majority of patients will have palliative care needs, defined as having a chronic, progressive and potentially fatal illness in an advanced or severe stage (malignant or non-malignant), no longer responding to curative/maintenance treatment or symptomatic, or both (e.g. lung/brain tumours or metastatic cancers, chronic obstructive pulmonary disease (COPD)).

We will evaluate interventions that are applicable in primary health care setting. Thus, interventions that require hospital multidisciplinary teams, devices only available in hospital care or that require hospitalization will be excluded.

QoL (primary outcome) will be assessed according to the score results of standardized and validated tools used to evaluate these domains (e.g. WHOQOL(16), QOLS(17), EQ-5D(18), SF-36(19), SF-12(19)).

As secondary outcomes, we will consider patients' symptoms relief (namely, pain, dyspnea, constipation, distress, depression and other important symptoms presented) and other patients' well-being measurements.

Search strategy

The search will be conducted in PsycInfo, EMBASE, MEDLINE, Cochrane and CINAHL from the start of indexing until December 2021. To identify ongoing clinical trials, we will also search in clinicaltrials.gov. No restriction on the language of publication will apply. See Supplementary file 1 for the search strategy in Medline. Searches in the other databases will be adapted accordingly.

The search will be complemented by a manual search of abstract books of relevant congresses and scientific meetings in the last five years, namely the *World Congress of the European Association for Palliative Care*, *International Congress on Palliative Care, WONCA Europe Conference*. Grey literature will also be accessed via Google Scholar search. Experts in this area will also be contacted to identify pertinent articles that may have not been identified by our query. Authors of retrieved articles will be contacted in case additional clarification is required.

When analyzing the search results, we will first screen the titles and abstracts of the articles found in the search. Then, all potential candidates will be selected by reading their full-text. Both phases will be independently carried out by two reviewers. Reasons for exclusion will be recorded. Disagreements will be solved by a third reviewer. Reviewer agreement in the selection process will be evaluated using agreement proportion and Kappa statistics.

Data extraction

The extraction of data from the studies will be independently carried out by two reviewers, with disagreements solved by a third reviewer. Customized data extraction tables will be created, piloted with five of the included studies and used to collect relevant data from all. To conduct data extraction, it will be used a standard data extraction form.

The following data will be extracted: general characteristics of the study (study design, sample size and setting), socio-demographic sample characteristics (gender, age and care context, e.g. home, care home, hospice, palliative care unit), main disease, main symptoms, the

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description of the intervention and comparator, methods of QoL assessment and other reported outcomes such as symptom relief or other measures of well-being.

Whenever possible, raw data will be extracted, allowing the calculation of the effect measure (such as QoL, symptom relief or other well-being measures). When data are ambiguous or missing from the published study, the corresponding author will be contacted.

Methodological quality assessment

We will evaluate the quality of the studies using the Cochrane risk-of-bias (RoB) 2.0 tool (20). Two reviewers will independently apply the tool to each study, with disagreements solved by a third reviewer. The grade of the evidence will be assessed using the Grading of Recommendations Assessment, Development and Evaluation (GRADE) (21) approach.

Data synthesis

We will undertake a narrative synthesis of the findings from the included studies, structured around the type of interventions and target population characteristics. Each intervention will be summarized in terms of its care context, content, and format, and resulting effects.

The primary outcome (QoL) will be described using standardised mean differences (SMD). Secondary outcomes will be described using SMD for continuous outcomes and risk ratio for categorical outcomes. The SMD will be categorised as small, medium and large based on the thresholds 0.2, 0.5 and 0.8, respectively, as suggested by Cohen's.(22)

In case of high heterogeneity (expected), subgroup analyses will be performed according to the type of intervention, main disease/main symptoms and care context.

Using the RevMan v.5.1® software, we will extract the data from the primary studies and analyze to decide whether it makes sense, according to heterogeneity and number of comparable studies, to perform a quantitative synthesis through meta-analysis to obtain an aggregate measure of QoL. Heterogeneity will be evaluated using the Cochran Q test (for a significance level of 0.05) and the I-squared statistic. In the absence of heterogeneity, we will use the Model of Fixed Effects. In the presence of slight to moderate heterogeneity (I-squared <40-50%), we will use the Random Effects Model. In case of severe heterogeneity (I-squared >40-50%), even in subgroup analyses, which impedes the accomplishment of an adequate quantitative synthesis, we will attempt to explain the existing variability.

If meta-analysis is performed, the pooled effect on QoL will be expressed as SMD. A P value of 0.05 will be used as the cut-off value to determine statistical significance and data will be presented as the estimated effect with 95% confidence intervals (CIs).

We will conduct a sensitivity analysis based on study quality and publication bias will be assessed by visual inspections of funnel plots.

We plan to start this review, be searching the databases, on April 1st and plan to have it completed for publication by August 31st**.**

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Patient and Public Involvement

Patients were not involved.

Discussion

This review will search, critically appraise and summarize the existing evidence on the effectiveness of non-pharmacological interventions applicable in primary care settings in improving the QoL of older adults with palliative care needs.

There are some studies assessing non-pharmacological interventions to improve the QoL of patients with palliative care needs (20-22). However, as far as we are aware, none summarizes the interventions that can be performed in primary care. Considering a holistic definition of palliative care, the involvement of primary care is essential, and its professionals need to be empowered with the best scientific evidence to deliver good quality care to their patients. By publishing the research protocol, we reinforce the clarity of the strategy and minimize the risk of bias, namely on selective outcome reporting.

Potential limitations of the review include the expected heterogeneity in terms of populations, interventions, care contexts and outcome measurement between the studies.

However, despite limitations, the review findings can be valuable and will be disseminated to help inform policy, service developments and professional training, with a view to improve care and thus the quality of life of the patients and their families.

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Footnotes

Contributorship Statement: CSC, JRM, FP and BG were involved in designing the study. MSM and CM were involved in the methodological section. CSC was involved in writing of the manuscript. All authors read and approved the final manuscript draft.

Author Statement: All authors have approved and contributed to the final written manuscript.

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Competing interests: None declared.

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Supplementary file 1

MEDLINE Query

Group 1

1. exp Terminal care/
2. exp Palliative care/
3. exp Terminally ill /
4. exp Hospices/
5. exp Death/
6. Palliat*.mp
7. Hospice*.mp
8. End of life.mp
9. Advanced or end stage or terminal adj4 disease* or illness*.mp
10. 1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 or 9

Group 2

11. exp Aged/
12. exp Old Age Assistance/
13. Old or aged or elderly.mp
14. 11 or 12 or 13

Group 3

15. exp Primary Health Care
16. exp General Practice
17. exp Family Practice
18. exp Community Health Centers
19. exp General Practitioners
20. exp Physicians, Family
21. exp Physicians, Primary Care
22. Primary care or family practice* or health center* or health centre*.mp
23. General practitioner* or GP* or family doctor*.mp
24. 15 or 16 or 17 or 18 or 19 or 20 or 21 or 22 or 23 or 24

Group 4

25. randomized controlled trial.pt
26. controlled clinical trial.pt
27. randomized.ti or randomized.ab
28. placebo.ti or placebo.ab
29. clinical trials as topic/
30. randomly.ti or randomly.ab
31. trial.ti
32. 25 or 26 or 27 or 28 or 29 or 30 or 31
33. animals.sh NOT humans.sh
34. 32 not 33

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Group 5

35. Exp "Quality of life"/
36. Quality of life.mp
37. 35 or 36

38. 10 and 14 and 24 and 34 and 37

To adapt to PsyncInfo, EMBASE, Medline, Cochrane and Cinahl search.

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Reporting checklist for protocol of a systematic review and meta analysis.

Based on the PRISMA-P guidelines.

Instructions to authors

Complete this checklist by entering the page numbers from your manuscript where readers will find each of the items listed below.

Your article may not currently address all the items on the checklist. Please modify your text to include the missing information. If you are certain that an item does not apply, please write "n/a" and provide a short explanation.

Upload your completed checklist as an extra file when you submit to a journal.

In your methods section, say that you used the PRISMA-Preporting guidelines, and cite them as:

Moher D, Shamseer L, Clarke M, Ghersi D, Liberati A, Petticrew M, Shekelle P, Stewart LA. Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols (PRISMA-P) 2015 statement. Syst Rev. 2015;4(1):1.

		Reporting Item	Page Number
Title			
Identification	#1a	Identify the report as a protocol of a systematic review	1
Update	#1b	If the protocol is for an update of a previous systematic review, identify as such	n/a
Registration			
	#2	If registered, provide the name of the registry (such as PROSPERO) and registration number	2
Authors			
Contact	#3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	2
Contribution	#3b	Describe contributions of protocol authors and identify the	7

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		guarantor of the review	
Amendments			
	#4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	n/a
Support			
Sources	#5a	Indicate sources of financial or other support for the review	7
Sponsor	#5b	Provide name for the review funder and / or sponsor	n/a
Role of sponsor or funder	#5c	Describe roles of funder(s), sponsor(s), and / or institution(s), if any, in developing the protocol	n/a
Introduction			
Rationale	#6	Describe the rationale for the review in the context of what is already known	4
Objectives	#7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	2,3,4
Methods			
Eligibility criteria	#8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	2,3
Information sources	#9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	3
Search strategy	#10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	10,11
Study records - data management	#11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	2,3,4
Study records -	#11b	State the process that will be used for selecting studies (such	2,3

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selection process		as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	
Study records - data collection process	#11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	3,4
Data items	#12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	3,4
Outcomes and prioritization	#13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	3,4
Risk of bias in individual studies	#14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	2,3,4
Data synthesis	#15a	Describe criteria under which study data will be quantitatively synthesised	3,4
Data synthesis	#15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I2, Kendall's τ)	3,4
Data synthesis	#15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	3,4
Data synthesis	#15d	If quantitative synthesis is not appropriate, describe the type of summary planned	n/a
Meta-bias(es)	#16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	3,4
Confidence in cumulative evidence	#17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	3,4

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